

The Strategy of Protectionism: Game Theory in International Trade

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Background

- February 2025: Trump Administration levies tariffs on Mexico, Canada, and China
- April 2025: 10% global tariff, 145% tariff on China, reciprocal tariffs on dozens of countries; foreign retaliation, negotiation
- How can this be modeled with game theory?

The Baseline Game

		China	
		T	NT
United States	T	1, 2	3.4, 1.5
	NT	0.75, 6	2.4, 5

Figure: The Baseline Game

- Payoffs based on the Trump Administration perspective of the effects of tariffs, in terms of 2024 GDP growth
- Nash Equilibrium: $(Tariff, Tariff)$, $Tariff$ is strictly dominant for both countries

Indefinite Prisoners' Dilemma

- Trade wars have not been the equilibrium
 - Half century of trade liberalization, so why tariff now?
- Indefinite game, where both countries play grim trigger (retaliatory tariffs)
- The discount rate of the U.S. has fallen below the threshold to sustain cooperation
 - $\delta \leq \frac{5}{12}$
- Some countries (E.U.) have not retaliated because their discount rate may still be high enough (for now)
 - $\delta \geq \frac{2}{3}$

A Game with Alternating Offers

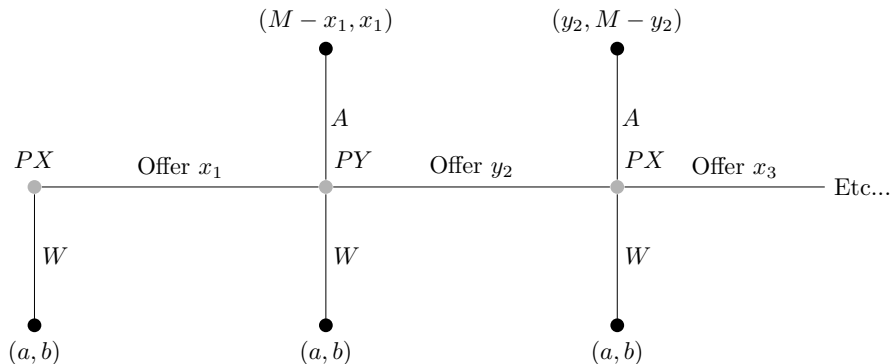


Figure: A Game With Alternating Offers

- The U.S. has made some trade deals
 - Rescinding tariffs on agricultural products (beef, coffee, & tea)
 - Let a foreign Adversary be PX and the U.S. be PY

A Game with Alternating Offers

U.S Discount Rate (δ_Y)	U.S Share (x^*)	Adversary's Share ($1 - x^*$)
0.10	0.0357	0.9643
0.2	0.0769	0.9231
0.30	0.1250	0.8750
0.4167	0.1923	0.8077
0.50	0.2500	0.7500
0.70	0.4375	0.5625
0.90	0.7500	0.2500

Table: The Share of The Gains from Trade, Holding δ_X Constant

- Assume $\delta_X = \frac{2}{3}$ and $\delta_Y = \frac{5}{12}$
- The lower discount rate of the U.S. harms their ability to negotiate
- More patience $\implies$ higher share of gains from trade